1)

```
package my_package;

parameter HEIGHT = 10;
parameter WIDTH  = 10;
parameter PERCENT_WHITE = 20;
typedef enum {BLACK, WHITE} colors_t;

class Screen;
    rand colors_t pixels [HEIGHT][WIDTH];
    constraint colors_c {
        foreach (pixels[i,j])
            pixels[i][j] dist {BLACK := 100-PERCENT_WHITE, WHITE := PERCENT_WHITE};
    }
    function void print_screen();
        foreach(pixels[i])
            $display("Row Number %0d = %0p",i,pixels[i]);
    endfunction
endclass

endpackage
```

```
import  my_package::*;

module image;
    Screen my_screen;
    initial begin
        my_screen=new();
        assert(my_screen.randomize());
        my_screen.print_screen;
    end
endmodule
```

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2)

```
package testing_pkg;

typedef enum {ADD, SUB, MULT, DIV} opcode_e;

class Transaction;

   rand opcode_e opcode;
   rand byte operand1;
   rand byte operand2;
   bit clk;

   covergroup CovCode @(posedge clk);
    operand1_cp: coverpoint operand1{
         bins max_neg = {-128};
         bins zero = {0};
         bins max_pos = {127};
         bins misc = default;
      }
    opcode_cp: coverpoint opcode{
         bins add_sub = {ADD, SUB};
         bins add_then_sub = (ADD=>SUB);
         illegal_bins no_div = {DIV};
      }

   endgroup

   function new();
      CovCode = new();
   endfunction

endclass // Transaction
endpackage : testing_pkg
```

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```
import testing_pkg::*;

module tb();
  Transaction tr = new;
  byte operand1, operand2, out;
  opcode_e opcode;
  bit clk, rst;
  initial begin
    clk = 0;
    forever begin
      #1 clk = ~clk;
      tr.clk = clk;
    end
  end

  //<DUT Instantiation>
  alu_seq DUT(operand1, operand2, clk, rst, opcode, out);

  initial begin
    rst = 1;
    @(negedge clk);
    rst = 0;
    repeat (32) begin // Run a few cycles
      assert(tr.randomize); // Create a transaction
      operand1 = tr.operand1;
      operand2 = tr.operand2;
      opcode = tr.opcode;
      @(negedge clk);
    end
    $stop();
  end // initial begin
endmodule
```

3)

```
covergroup CovCode @(posedge clk);
 operand1_cp: coverpoint operand1{
      bins max_neg = {-128};
      bins zero = {0};
      bins max_pos = {127};
      bins misc = default;
   }
 opcode_cp: coverpoint opcode{
      bins add_sub = {ADD, SUB};
      bins add_then_sub = (ADD=>SUB);
      illegal_bins no_div = {DIV};
   }

  opcode_operand1: cross opcode_cp, operand1_cp {
      option.weight = 5;
      option.cross_auto_bin_max = 0;
      bins max_pos_add_or_sub = binsof(operand1_cp.max_pos) && binsof(opcode_cp.add_sub);
      bins max_neg_add_or_sub = binsof(operand1_cp.max_neg) && binsof(opcode_cp.add_sub);
   }

endgroup
```

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